

Overview and Essentiality:

Boron is a chemical element that is present naturally in many foods and is available as a dietary supplement. It is not currently classified as an essential nutrient for humans because a clear biological function has not been established. However, evidence suggests that boron may influence the metabolism of other minerals, including calcium, magnesium, and phosphorus, and may play a role in bone maintenance and steroid hormone activity.

Dietary Sources of Boron:

Plant foods are the primary dietary sources of boron. High concentrations of boron are found in fruits, fruit juices, nuts, and legumes:

- Fruits: Avocados, red grapes, raisins, peaches, apples, and pears.
- Nuts and Legumes: Peanuts, almonds, hazelnuts, kidney beans, and lentils.
- Beverages: Wine, cider, and beer also contribute to boron intake in adult diets.

Animal-derived foods (meat, poultry, fish, and dairy) are relatively low in boron. Most adults in the United States consume approximately 1 mg of boron per day through a standard diet.

Biological Roles:

Studies indicate that boron interacts with vitamin D, steroid hormones (such as estrogen and testosterone), and cellular antioxidant defenses. Boron supplements are commercially available as boron citrate, boron glycinate, boron aspartate, and calcium fructoborate.

Tolerable Upper Intake Levels (UL) for Boron:

The Food and Nutrition Board (FNB) of the National Academies established Tolerable Upper Intake Levels (ULs) for boron for the general population. The UL is the highest level of daily intake that is likely to pose no risk of adverse health effects in almost all individuals:

Age Group and Tolerable Upper Intake Level (UL):

- Birth to 12 months: Not established (intake should be from breast milk, formula, and food)
- Children 1-3 years: 3 mg/day
- Children 4-8 years: 6 mg/day
- Children 9-13 years: 11 mg/day
- Adolescents 14-18 years: 17 mg/day
- Adults 19 years and older: 20 mg/day (including pregnant and lactating females)

Health Risks of Excessive Boron:

The UL for boron was established based on animal studies showing adverse effects on male reproduction and embryonic development. In humans, acute ingestion of high doses of boron (such as boric acid) can cause nausea, gastrointestinal discomfort, vomiting, diarrhea, dermatitis, and lethargy. Chronic excessive intake above the 20 mg/day adult upper limit is not recommended.